UCC 301.COM 422 L8

RESEARCH METHODOLOGY

Research methodology are the systematic procedures by which the researcher starts from the initial identification of the problem to its final conclusions ensuring that the research is carried out in a scientific and valid manner. It includes the research paradigm, the research approach, design, population sampling, data collection, data analysis and the ethical issues in the research

Research paradigms

Paradigm indicate a pattern or model or typical example including cultural themes, worldviews, Ideologies and mindsets. They can also be described as are general framework or viewpoints: literally points from which to view. They provide ways of looking at life and are grounded in sets of assumptions about the nature of reality.

A research paradigm is the set of common beliefs and agreements shared between scientist about how problems should be understood and addressed. Research paradigms can be characterized by the way Scientists respond to three basic questions: ontological, epistemological and methodological questions.

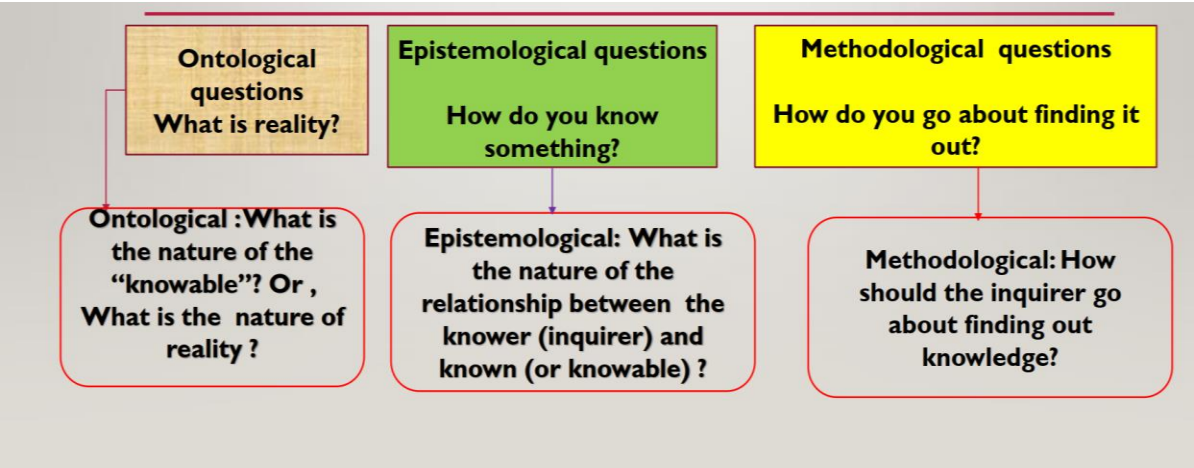
Ontology is the study of beings or their being: What is reality?

Epistemology is the study of knowledge: How we know?

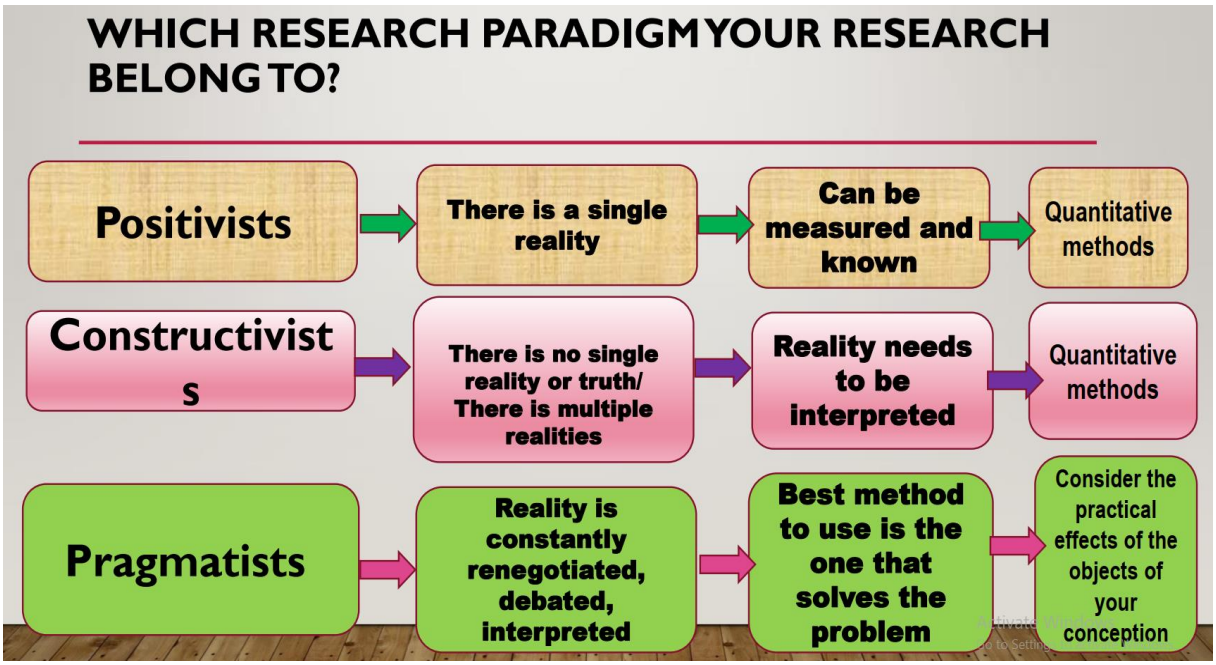
Logic is the study of valid reasoning: How we reason?

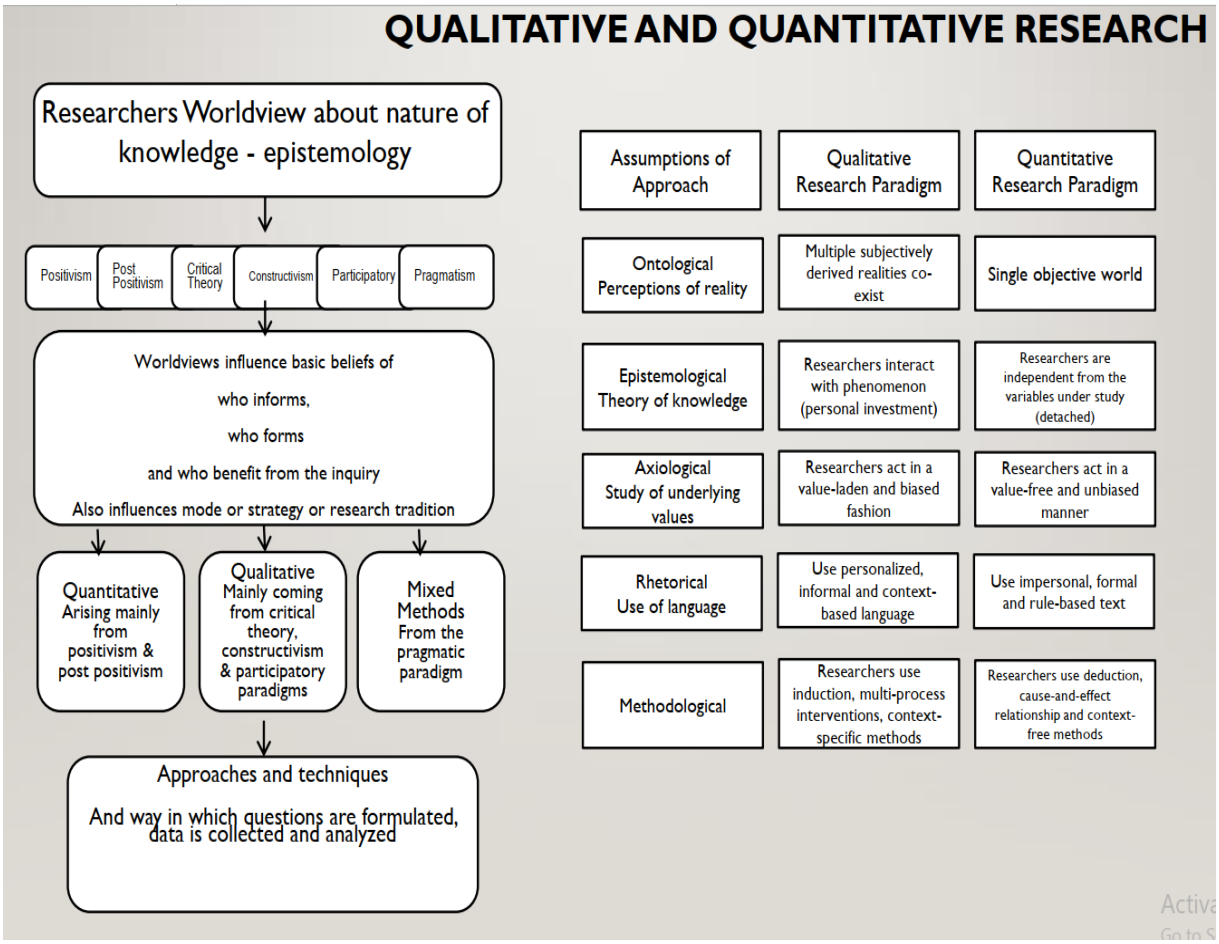
Ethics is the study of right and wrong: How we should act?

Phenomenology is the study of our experience: How we experience?



The major research paradigms are: Positivism; Post-positivism; Critical theory (ideology); Constructivism; pragmatists


WHICH RESEARCH PARADIGM YOUR RESEARCH BELONG TO?
Positivists
There is a single reality
Can be measured and known
Quantitative methods
Constructivists
There is no single reality or truth/ There is multiple realities
Reality needs to be interpreted
Quantitative methods
Pragmatists
Reality is constantly renegotiated, debated, interpreted
Best method to use is the one that solves the problem
Consider the practical effects of the objects of your conception


QUALITATIVE AND QUANTITATIVE RESEARCH
Researchers Worldview about nature of knowledge - epistemology
Positivism
Post Positivism
Critical Theory
Constructivism
Participatory
Pragmatism
Worldviews influence basic beliefs of
who informs,
who forms
and who benefit from the inquiry
Also influences mode or strategy or research tradition
Quantitative Arising mainly from positivism & post positivism
Qualitative Mainly coming from critical theory, constructivism & participatory paradigms
Mixed Methods From the pragmatic paradigm
Approaches and techniques
And way in which questions are formulated, data is collected and analyzed
Assumptions of Approach
Qualitative Research Paradigm
Quantitative Research Paradigm
Ontological Perceptions of reality
Multiple subjectively derived realities co-exist
Single objective world
Epistemological Theory of knowledge
Researchers interact with phenomenon (personal investment)
Researchers are independent from the variables under study (detached)
Axiological Study of underlying values
Researchers act in a value-laden and biased fashion
Researchers act in a value-free and unbiased manner
Rhetorical Use of language
Use personalized, informal and context-based language
Use impersonal, formal and rule-based text
Methodological
Researchers use induction, multi-process interventions, context-specific methods
Researchers use deduction, cause-and-effect relationship and context-free methods

Research approach

Read lecture 2 on qualitative, quantitative and mixed methods

Research designs

A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure. The research design refers to the overall strategy that you choose to integrate the different components of the study in a coherent and logical way, thereby, ensuring you will effectively address the research problem; it constitutes the blueprint for the collection, measurement, and analysis of data. There are numerous types of research design that are appropriate for the different types of research projects. The choice of which design to apply depends on the nature of the problems posed by the research aims. Each type of research design has a range of research methods that are commonly used to collect and analyses the type of data that is generated by the investigations.

Note that your research problem determines the type of design you should use, not the other way around

Features of a good design

- The design which minimizes bias and maximizes the reliability of the data collected and analyzed is considered a good design.
- The design which gives the smallest experimental error is supposed to be the best design in many investigations. Similarly,
- A design which yields maximal information and provides an opportunity for considering many different aspects of a problem is considered most appropriate and efficient design in respect of many research problems.

Qualitative Research designs

There are five major types of qualitative research: phenomenology, ethnography, case study research, grounded theory, and historical research. All of the approaches are similar in that they are qualitative approaches. Each approach, however, has some distinct characteristics and tends to have its own roots and following.

Phenomenology– a form of qualitative research in which the researcher attempts to understand how one or more individuals experience a phenomenon. For example, you might interview 20 widows and ask them to describe their experiences of the deaths of their husbands.

Ethnography– is the form of qualitative research that focuses on describing the culture of a group of people. Note that a culture is the shared attitudes, values, norms, practices, language, and material things of a group of people. For an example of ethnography, you might decide to go and live in a Mohawk community and study the culture and their educational practices.

Case study research– is a form of qualitative research that is focused on providing a detailed account of one or more cases. For an example, you might study a classroom that was given a new curriculum for technology use.

Grounded theory– is a qualitative approach to generating and developing a theory from data that the researcher collects. For an example, you might collect data from parents who have pulled their children out of public schools and develop a theory to explain how and why this phenomenon occurs, ultimately developing a theory of school pull-out.

Historical research– research about events that occurred in the past. An example, you might study the use of corporeal punishment in schools in the 19th century.

Quantitative Research Design

Descriptive research

Descriptive studies aim to explore and describe a phenomenon without any manipulation. Data collected methods such as observation, interviews and questionnaires are used but unlike qualitative studies the data collected is numerical often using standardized tools and measures. This type of study allows for possible relationships to be sought and the development of hypotheses that can be tested in future quantitative studies. Often, simple descriptive statistics such as frequency, mean, mode or median are used to report data from descriptive studies.

Correlational research

Correlational research aims to explore relationships / associations between given variables; the correlation or extent of an association is reported numerically known as a correlation coefficient. A positive correlation describes findings when both variable; a negative correlation exists when one variable increases and the other decreases. In statistical reporting, a perfect positive correlation is represented by the value 1, while 0 indicates no correlation and -1 indicates a perfect negative

correlation. Various tools are used to collect data in correlational research including validated questionnaire or scales.

Quasi-experimental research

The aim of quasi-experimental research is to examine causal relationships between variables. These types of studies are considered quasi-experimental since they lack the element randomization seen in experimental studies.

Experimental Research

This type of research, often considered to be the gold standard, is objective, systematic and highly controlled; the double blind, randomized controlled trial (RCT) being a perfect example. This type of research allows causality between independent and dependent variables to be examined under highly controlled conditions. A range of statistical tests might be used in these types of studies depending on the data collected.

Simulation

Simulation involves devising a representation in a small and simplified form (model) of a system, which can be manipulated to gauge effects. It is similar to experimental design in the respect of this manipulation, but it provides a more artificial environment in that it does work with original materials at the same scale. Models can be mathematical (number crunching in a computer) or physical, working with two- or three-dimensional materials. The performance of the model must be checked and calibrated against the real system to check that the results are reliable. Simulation enables theoretical situations to be tested – what if?

Surveys

Surveys involve the collection and analysis of information from a group of people drawn from a specifically targeted group of interest. Surveys of inanimate objects can also be conducted.

A survey is the planned collection of information in a standardized form that aims to give an overall perspective to a field. It permits a number of variables to be examined, which can be used to test several hypotheses and can be analyzed in a number of different ways. Surveys are easily replicated when standardized sampling and analysis techniques are used. The accuracy of responses will vary

according to how well respondents understand the questions and the issues, how much bias exists, and whether extraneous variables are distorting results.

Population and sampling

Definition - a complete set of elements (persons or objects) that possess some common characteristic defined by the sampling criteria established by the researcher

Composed of two groups - target population & accessible population

Target population (universe): The entire group of people or objects to which the researcher wishes to generalize the study findings

Accessible population: the portion of the population to which the researcher has reasonable access; may be a subset of the target population

Target population

Target population is the specific complete group relevant to the research project. To define the population a researcher must specify the unit, being sampled, the geographical location and temporal boundaries of the population.

Sampling

Sampling refers to the systematic selection of a limited number of elements (persons, objects or events) out of a theoretically specified population of elements, from which information will be collected.

A sample is a smaller (but hopefully representative) collection of units from a population used to determine truths about that population

Sampling frame

A researcher operationalizes a population by developing a specific list that closely approximate all the elements in the population. This list is referred to as sampling frame.

A sampling frame is the list of elements from which the sample may be drawn. A simple example could be listing of all college students meeting the criteria of target population and who are enrolled on the specified date.

Sampling Frame Error

A sampling frame error occurs when certain sample elements are excluded or when the entire population is not accurately represented in the sampling frame. The error that occurs when certain sample elements are not listed or available and are not represented in the sampling frame.

Sampling Unit

A sampling unit is that element or set of elements considered for selection in some stage of sampling. Sampling may be done in single stage or in multiple stages. In a simple, single stage sample, the sampling units are the same as the elements. In more complex samples, however, different levels of sampling units may be employed.

SAMPLING TECHNIQUES

PROBABILITY SAMPLING TECHNIQUES

In probability sampling every element in the population has a known nonzero probability of selection. It involves the following techniques

Random Sampling: Each person in the universe has an equal probability of being chosen for the sample and every collection of persons of the sample has an equal probability of becoming the actual sample. Example: in an alphabetical list of all students enrolled at KaRU, each student is given a number ascending from 1, and 400 students are selected using a table of random numbers.

Systematic Sampling: A sample constructed by selecting every kth element in the sampling frame. Systematic sampling is more practical in that it is less work and thus provides more information per dollar. It also may reduce error. (The more complex the method, the greater opportunity for error.) Example: in a list of all students enrolled at KaRU, each 100th student, starting with the randomly chosen 205th, is selected. Later it turns out that every other student in the list is female (and the entire sample female.

Stratified Random Sampling: Is obtained by separating the population elements into overlapping groups, called strata, and then selecting a simple random sample from within each stratum. Example: Rank order: full professor, associate professor, etc. After this is done a random or systematic sample is drawn within each group.

Cluster Sampling: A simple random sample in which each sampling unit is a collection or cluster, or elements. For example, an investigator wishing to study students might first sample

groups or clusters of students such as classes or dormitories, and then select the final sample of students from among clusters. Also called area sampling. Advantage: saves time and money. Disadvantage: error.

NON PROBERABILITY

In non-probability sampling the probability of any particular element of the population being chosen is unknown. The selection of units in non-probability sampling is quite arbitrary, as researchers rely heavily on personal judgment. These techniques include:

Convenience Sampling: The investigator chooses the closest live persons as respondents "Captive audience" sampling. (E.g., using a class of Com 422 students).

Quota Sampling: Equivalent to a stratified sample with the added requirement that each stratum is generally represented in the sample in the same proportion as in the entire population. (Example: 60% male students and 40% Female - select a sample of the two)

Dimensional Sampling: A multidimensional form of quota sampling. One has to specify all dimensions (variables) or interest on the population and then to make sure that every combination of these dimensions are represented by at least one case. This method is designed for studies in which only a small sample is desired so that each case drawn can be studied in more detail than is possible in a large-scale study.

Purposive Sampling: The researcher uses his or her own judgment about which respondents to choose, and picks those who best meets the purposes of the study.

Snowball Sampling: Has achieved increased use in recent years; particularly by researchers conducting observational research and in community studies. Conducted in stages. In the 1st stage a few persons having the requisite characteristics are identified and interviewed. These persons are used as informants to identify others who qualify for inclusion in the sample.

The second stage involves interviewing these persons who in tum lead to still more persons who can be interviewed in the 3rd stage, etc. The term "snowball" stems from the analogy of a snowball, which begins small but becomes bigger and bigger and rolls downhill. Also called "chain referral sampling." particularly useful in deviant studies and subcultures (drug addicts, etc.) Sample Size

- Depends on the size of the population to be sampled.

- Lack of adequate representation can be referred to as sampling error.

Sample Bias

A good sample is **representative**. This means that each sample point represents the attributes of a known number of population elements.

Bias often occurs when the sample does not accurately represent the population. The bias that results from an unrepresentative sample is called **selection bias**. Some common examples of selection bias are described below.

- **Undercoverage**. Undercoverage occurs when some members of the population are inadequately represented in the sample. A classic example of undercoverage is the *Literary Digest* voter survey, which predicted that Alfred Landon would beat Franklin Roosevelt in the 1936 presidential election. The survey sample suffered from undercoverage of low-income voters, who tended to be Democrats.

  How did this happen? The survey relied on a convenience sample, drawn from telephone directories and car registration lists. In 1936, people who owned cars and telephones tended to be more affluent. Undercoverage is often a problem with convenience samples.

- **Nonresponse bias**. Sometimes, individuals chosen for the sample are unwilling or unable to participate in the survey. Nonresponse bias is the bias that results when respondents differ in meaningful ways from nonrespondents. response bias.

- **Voluntary response bias**. Voluntary response bias occurs when sample members are self-selected volunteers, as in voluntary samples. An example would be call-in radio shows that solicit audience participation in surveys on controversial topics (abortion, affirmative action, gun control, etc.). The resulting sample tends to overrepresent individuals who have strong opinions.

Bias Due to Measurement Error

A poor measurement process can also lead to bias. Eg In survey research, the measurement process includes the environment in which the survey is conducted, the way that questions are asked, and the state of the survey respondent.

Response bias refers to the bias that results from problems in the measurement process. examples of response bias are:

- **Leading questions**. The wording of the question may be loaded in some way to unduly favor one response over another. For example, a satisfaction survey may ask the respondent to indicate where she is satisfied, dissatisfied, or very dissatified. By giving the respondent one response option to express satisfaction and two response options to express dissatisfaction, this survey question is biased toward getting a dis-satisfied response.

- **Social desirability**. Most people like to present themselves in a favorable light, so they will be reluctant to admit to unsavory attitudes or illegal activities in a survey, particularly if survey results are not confidential. Instead, their responses may be biased toward what they believe is socially desirable.

Sampling Error and Survey Bias

A survey produces a sample statistic, which is used to estimate a population parameter. If you repeated a survey many times, using different samples each time, you might get a different sample statistic with each replication. And each of the different sample statistics would be an estimate for the *same* population parameter.

If the statistic is unbiased, the average of all the statistics from all possible samples will equal the true population parameter; even though any individual statistic may differ from the population parameter. The variability among statistics from different samples is called **sampling error**.

Increasing the sample size tends to reduce the sampling error; that is, it makes the sample statistic less variable. However, increasing sample size does not affect survey bias. A large sample size cannot correct for the methodological problems that produce bias.